

Data Cleaning and Preparation Report

Summary of data loading, missing-value handling, duplicate removal, data-type correction, and date verification.

Metric	Result
Source file	Dataset for Data Analytics.xlsx
Original DataFrame shape	(1200, 14)
Cleaned DataFrame shape	(891, 14)
Missing CouponCode values before cleaning	309
Duplicate rows found	None
Date range	2023-01-01 to 2025-06-30

1. Initial Data Loading and Missing Value Handling

Item	Details
Action	Loaded the Excel file Dataset for Data Analytics.xlsx into a pandas DataFrame.
Initial state	The original DataFrame had a shape of (1200, 14).
Missing values	Before cleaning, the CouponCode column contained 309 missing values.
Cleaning action	All rows containing any missing values were dropped using df.dropna().
Result	The cleaned DataFrame, df_cleaned, has a shape of (891, 14), and all missing values were successfully removed.

2. Duplicate Removal

Item	Details
Action	Duplicate rows were checked for and removed from the df_cleaned DataFrame.
Result	No duplicate rows were found. The DataFrame shape remained (891, 14), confirming the uniqueness of the remaining records.

3. Data Type Correction

Item	Details
Action	The data types of several columns were inspected and corrected.
Date	Successfully converted to datetime64[ns].
Quantity and ItemsInCart	Converted to Int64, an integer type capable of handling NaN values.
UnitPrice and TotalPrice	Confirmed as float64.

Item	Details
Result	The data types are now appropriate for analysis. Date is correctly formatted for time-series operations, and numerical columns are set for calculations.

4. Date Column Verification

Item	Details
Action	The Date column was verified after type conversion.
Result	The Date column is confirmed to be datetime64[ns]. The minimum date is 2023-01-01 and the maximum date is 2025-06-30.

Overall status: The dataset was cleaned successfully, duplicate-free, correctly typed, and verified for valid date coverage.